

Byte Pair Encoding (BPE)

This sample uses a practice corpus that is different from the corpora used in the assignment. It illustrates the expected method of calculation and presentation.

Assignment Question

Using the following artificial training corpus. The number beside each word is its corpus frequency. Append the end-of-word marker _ to every word. Count adjacent symbol pairs using the word frequencies. Apply the stated tie-breaking rule and complete three BPE merges.

Word	Frequency
walk	5
walking	3
walker	2
walked	2

Tie-breaking rule: if two or more pairs have the same maximum frequency, select the pair that appears first when the corpus is scanned in the listed word order, from left to right.

Sample Answer

1. Initial symbol representation

5 w a l k _
3 w a l k i n g _
2 w a l k e r _
2 w a l k e d _

The frequency is not itself a token. It tells us how many times that word occurs in the training corpus.

2. Iteration 1

Initial weighted pair counts are:

Adjacent pair	Weighted frequency
w a	12
a l	12
l k	12
k _	5
k i	3
i n	3
n g	3
g _	3
k e	4
e d	2
d _	2
e r	2
r _	2

The maximum frequency is 12, shared by w a, a l and l k. By the tie-breaking rule, w a is encountered first.

Merge: w + a -> wa

Updated corpus:

5 wa l k _
3 wa l k i n g _
2 wa l k e d _
2 wa l k e r _

3. Iteration 2

The adjacent pairs are counted again after the first merge:

Adjacent pair	Weighted frequency
wa l	12
l k	12
k _	5
k i	3
i n	3
n g	3
g _	3
k e	4
e d	2
d _	2
e r	2
r _	2

The maximum frequency is again 12. The tied pairs are wa l and l k. The pair wa l occurs first under the stated rule.

Merge: wa + l -> wal

Updated corpus:

5 wal k _
3 wal k i n g _
2 wal k e d _
2 wal k e r _

4. Iteration 3

Recalculate the pair frequencies:

Adjacent pair	Weighted frequency
wal k	12
k _	5
k i	3
i n	3
n g	3
g _	3
k e	4
e d	2
d _	2
e r	2
r _	2

The pair for m has the highest frequency, 12.

Merge: wal + k -> walk

Updated corpus after three merges:

5 walk _
3 walk i n g _
2 walk e d _
2 walk e r _

5. Summary of the three merges

Iteration	Most frequent pair	Weighted frequency	New token
1	w a	12	wa
2	wa l	12	wal
3	wal k	12	walk